

Software Requirements Specification (Updated)

Online Class Recording & AI Evaluation Platform

Executive Summary

This system provides online classes, attendance tracking, recording management, transcript generation, AI-powered post-class evaluation, and automated reporting. The platform does not perform real-time monitoring. All analysis occurs after class completion.

Project Objectives

Provide a scalable online learning platform, generate AI-powered class evaluation reports, ensure teaching quality, detect policy violations, and maintain complete class records.

System Scope

The platform supports scheduling, video classes, attendance, recording, transcript generation, AI analysis, report generation, and administrative review.

User Roles

Admin, Teacher, Student, Compliance Reviewer.

High-Level Workflow

Class Starts → Recording → Google Drive Upload → Audio Extraction → Speech-to-Text → AI Analysis → Report Generation → Admin Review.

Functional Requirements

- FR-01 Authentication & Authorization
- FR-02 Course Management
- FR-03 Student Enrollment
- FR-04 Class Scheduling
- FR-05 Live Video Classes via LiveKit
- FR-06 Attendance Tracking
- FR-07 Session Recording
- FR-08 Google Drive Storage
- FR-09 Transcript Generation
- FR-10 AI Transcript Analysis
- FR-11 Automated Report Generation
- FR-12 PDF Export
- FR-13 Admin Review Dashboard

Use Cases

- UC-01 Login
- UC-02 Create Course
- UC-03 Schedule Class
- UC-04 Start Class
- UC-05 Join Class
- UC-06 Record Class
- UC-07 Upload Recording

UC-08 Generate Transcript
UC-09 Analyze Transcript
UC-10 Generate AI Report
UC-11 Review Reports
UC-12 Export PDF

Technology Stack

Frontend: Next.js 15
Backend: NestJS
Database: PostgreSQL + Prisma
Video Platform: LiveKit
Queue: Redis + BullMQ
Storage: Google Drive API
Speech-to-Text: Whisper / Google STT
AI Analysis: Gemini / OpenAI
Reverse Proxy: Nginx

Frontend Requirements

Role-based dashboards, responsive design, class scheduling interface, recording access, report viewing, transcript viewer, PDF export, dark mode support.

Backend Modules

Authentication Module
User Management Module
Course Module
Enrollment Module
Class Scheduling Module
Attendance Module
Recording Module
Google Drive Module
Transcript Module
AI Analysis Module
Report Module
Notification Module

Database Design

User
Course
Enrollment
ClassSession
Attendance
Recording
TranscriptSegment
AIReport
Violation
Notification
AuditLog

Attendance Requirements

Track:
- Join Time
- Leave Time
- Total Attendance Duration
- Student Attendance Percentage
- Teacher Attendance Duration

Recording Requirements

Automatically record every class.
Store metadata.
Upload recordings to Google Drive.
Maintain recording statuses:
PROCESSING
UPLOADING
READY

FAILED

Google Drive Requirements

Create folders automatically.
Upload recordings.
Save file IDs and shareable URLs.
Retry failed uploads.
Remove local temporary files after success.

Speech-to-Text Pipeline

Extract audio using FFmpeg.
Convert audio to text.
Store transcript with timestamps.
Support large recordings.
Support multiple languages.

AI Analysis Requirements

Analyze transcript after class completion.

Detect:

- Phone numbers
- Email addresses
- WhatsApp numbers
- Social media sharing
- Off-topic discussion
- Inappropriate language
- Missing educational content
- Excessive non-academic discussion

Generate:

- Risk score
- Teaching quality score
- Topic relevance score
- Summary
- Recommendations

AI Report Structure

Class Information
Teacher Information
Attendance Summary
Class Duration
Transcript Summary
Main Topics Covered
Off-topic Analysis
Contact Sharing Detection
Compliance Findings
Teaching Quality Assessment
Student Engagement Assessment

Recommendations

Final Score

Google Drive Recording Link

Admin Dashboard

- View Reports
- Search Reports
- Filter Reports
- Review Violations
- Download PDFs
- Access Recordings
- View Attendance Statistics

Teacher Dashboard

- Upcoming Classes
- Previous Classes
- Recording History
- Generated Reports
- Attendance Statistics

Student Dashboard

- Upcoming Classes
- Join Classes
- Attendance History
- Recording Access

Security Requirements

- JWT Authentication
- RBAC Authorization
- TLS Encryption
- Rate Limiting
- Audit Logging
- Secure Secret Storage
- Data Encryption at Rest

Performance Requirements

- Support 1000+ concurrent users.
- Dashboard load time under 2 seconds.
- Report generation under 5 minutes after class completion.

Docker Architecture

- Next.js Container
- NestJS Container
- PostgreSQL Container
- Redis Container
- LiveKit Container
- Worker Container
- Nginx Container

CI/CD Pipeline

GitHub Actions

Linting

Unit Tests

Integration Tests

Docker Build

Security Scan

Staging Deployment

Production Deployment

Monitoring & Logging

Prometheus
Grafana
Loki
Health Checks
Error Tracking

Deployment Strategy

Development Environment
Staging Environment
Production Environment

Automated migrations.
Automated backups.
SSL certificates.
Monitoring setup.

Testing Strategy

Unit Testing
Integration Testing
E2E Testing
Load Testing
Security Testing
User Acceptance Testing

Development Roadmap

Phase 1: Foundation & Authentication
Phase 2: Scheduling & Enrollment
Phase 3: LiveKit Integration
Phase 4: Recording & Google Drive
Phase 5: Attendance Tracking
Phase 6: Speech-to-Text Pipeline
Phase 7: AI Analysis Engine
Phase 8: Report Generation
Phase 9: Dashboard & Exports
Phase 10: Production Deployment

Acceptance Criteria

Admin can create classes.
Teachers can host classes.
Students can attend classes.
Recordings upload successfully.
Transcripts generate automatically.
AI reports generate after every class.
Reports are visible to admins.
PDF exports work.

System deploys successfully to production.